D07 · Warehouse Operations & WMS

05 · Value-added services (VAS) & postponement in the warehouse

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

1 WHAT IT IS

Value-added services (VAS) are customization steps performed in the warehouse — labeling, kitting, repackaging, serialization, personalization; postponement delays final product differentiation until the last responsible point in the chain.

2 WHY IT MATTERS

Holding product in generic form and configuring it late lets one inventory serve many variants, which reduces total stock and obsolescence **(Feitzinger and Lee, 1997 — Harvard Business Review, on postponement at Hewlett-Packard)**. **The condition is quality and speed: VAS must be integrated into the WMS and controlled like manufacturing**, or it creates errors and traceability gaps.

3 HOW IT IS DONE

1. **Find the postponement candidates.** Products with many variants differing only in label, pack or configuration.
2. **Run make vs. buy.** Compare warehouse VAS cost and speed with supplier or plant alternatives.
3. **Integrate VAS into the WMS.** Work orders, component reservations and scan verification at each step.
4. **Measure quality and cost per unit.** Error rate, rework and cost per VAS unit, with standards like any production line.

IN PRACTICE

Postponement can reduce total inventory 20–40% in chains with high SKU diversity

A product with 4 regional variants (label in 4 languages) can be managed as 1 generic SKU in the central warehouse and differentiated at the regional DC or POS. Required inventory is significantly lower than 4 independent SKUs through demand aggregation.

Integrate VAS into the WMS flow from design — don’t add it as a parallel process

VAS “outside the WMS” (operator assembles the kit without the WMS knowing) generates inventory discrepancies, lack of traceability, and difficulty measuring productivity and quality.

The make vs. buy analysis for VAS must include flexibility and response speed

A VAS performed in our warehouse can be executed with 24 hours of notice. The same VAS performed by the supplier requires 4 weeks. Response speed is a benefit of internal VAS that a purely financial cost analysis may underestimate.

Define VAS quality standards with the same criteria as finished product

A kit incorrectly assembled in the warehouse is a product quality defect — not a warehouse error. VAS quality standards must be as rigorous as the manufacturing process standards.

ILLUSTRATIVE CASE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case: Kitting implementation in warehouse — electronics company, 12 promotional kits — The company launches 12 new promotional kits that must be assembled in the DC (1 tablet + 1 keyboard + 1 case) before dispatch.

Kitting process element	Design	KPI achieved
Dedicated kitting station with WMS integration	3 stations · WMS guides assembly and scanner-verifies each component	Kitting error rate: 0.08% (1 error in 1,250 kits) vs. 2.4% without scanner

Kitting process element	Design	KPI achieved
Kitting productivity	Standard: 28 kits/hr per operator	Achieved: 31 kits/hr (111% of standard in month 2)
Kit traceability (each component's lot)	WMS captures each component's serial number in the kit	Full traceability: 100% of kits with identified components
Kit component inventory management	WMS inventory reservation for active kit components	0 kitting process stoppages due to missing components in months 2 and 3

Result: Kitting implemented in 4 weeks. Error rate: 0.08% vs. 2.4% without WMS guidance. Productivity: 31 kits/hr. Integrating kitting into the WMS was the most important decision — manual kitting outside the WMS generated component errors and traceability loss.

HOW IT IS MEASURED

VAS Quality Rate % (VAS executed without error or rework)

VAS Quality Rate %(VAS units executed correctly without rework / Total VAS units processed) × 100

Benchmark: >99.5% for kitting · >99.9% for labeling and serialization A VAS quality rate <98% indicates a VAS process with insufficient quality control — generating errors that reach the customer and are more costly to correct after the fact than to prevent at the source.

VAS Cost per Unit (cost of each VAS unit executed)

VAS Cost per UnitTotal VAS operation cost (labor + materials + overhead) / Total VAS units processed

Benchmark: Benchmark vs. the cost of the same VAS performed by the supplier (make vs. buy) · Declining trend indicates process efficiency improvement In-warehouse VAS only makes financial sense if the cost is lower than performing it at the supplier, plant, or by the customer — and if the flexibility it generates has strategic value.

WHAT YOU WOULD USE

VAS & Postponement

SAP EWM — Warehouse Task Management — Module: VAS Integration + Kitting + LabelingSAP EWM manages VAS processes (kitting, labeling, repackaging) within the WMS flow, with full traceability and component inventory management.

Manhattan Associates WMS — Value-Added Services — Module: VAS Workflows + Quality ControlManhattan Associates VAS module with configurable workflows for each VAS type and scanner confirmation at each process step.

4 SO WHAT

Differentiate late, control tightly. Postpone where variants share a generic base, integrate VAS into the WMS, and hold it to manufacturing-grade quality standards.

From the SCRA v2.2 reference architecture · mrsupplychain.ai · © 2026 Ismael López · Mr. Supply Chain® México